The Spencer–Huggins No-Decompression Tables: Reconstruction, Numerical Audit, and Open Questions

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Abstract

We reconstruct the six-compartment model underlying Huggins's 1981 no-decompression tables, using Spencer's exposure limits as tabulated by Huggins. All six calibrated surfacing limits are recovered; 70 of 72 intermediate pressures agree at the printed precision. We distinguish the mathematical model from the three historical charts, preserve their discrepancies, and identify a nearly additive structure in the surface-interval chart: 63 of 66 nontrivial cross-cell predictions are exact and three differ by one minute. An exact rational calculation verifies 5,988 compartment inequalities for 998 specified first-dive/surface-interval combinations. The original model's six fixed compartments are irredundant, but minimality among refitted models is left open. Probable printing errors and possible undocumented table conventions are labeled as explanations rather than established historical facts. The accompanying software and data make each numerical assertion reproducible. No physiological validation or complete recovery of the unpublished calculator program is claimed.

1 What is reconstructed

Huggins's report [1] provides a compact test case for reconstructing a historical dive table from its stated mathematical ingredients. The purpose of this note is to answer the questions those ingredients determine, identify discrepancies precisely, and leave undocumented historical choices open. Table 1 summarizes the results.

Table 1: Audit results. Numerical agreement, model-bound compliance, and source transcription are different assertions.

Object checked	Result
Calibrated surfacing limits	6 of 6 reproduced
Intermediate compartment pressures	70 of 72 match printed precision
First-dive numerical entries	120 of 127 satisfy nominal group bounds
Finite residual-time entries	169 of 179 match a stated inverse candidate
Surface-interval transitions	91 of 91 exceed the derived box bound
Nontrivial additive predictions	63 exact; 3 differ by one minute
First-dive/surface-interval inequalities	5,988 verified by rational bounds

Scope. The inputs are Spencer's limits *as printed by Huggins*, not Spencer's raw Doppler observations. A square first dive means an initially surface-equilibrated state held at one depth, with

instantaneous pressure changes. The certificate below concerns this model and one subsequent surface interval. Huggins explicitly distinguished mathematical development from experimental validation [1, disclaimer and p. 6]. This note and its software are for research, not dive planning, and make no claim about acceptable decompression-sickness risk.

1.1 Source witnesses and preserved readings

In the NOAA copy of the 1981 report, printed pages 1–6 are PDF pages 4–9. Page 1 lists the empirical limits; pages 4–5 describe the kinetic formula, calibration, and group percentages; pages 2–3 contain the first-dive, surface-interval, and residual-nitrogen-time charts. The scan clips columns J and K of the first two charts. Those columns alone are recovered from the author's 1987 reproduction, Appendix D, printed page 99 [2]. The transcription therefore uses two identified witnesses. Legible 1981 entries take precedence.

The fixture retains numerical entries, dashes, right arrows, and blanks separately. A blank or dash is not automatically interpreted as infinity. Huggins later explains that the arrows direct the reader rightward to obtain a group designator and that the groups account for all six compartments [2, p. 13]. These are documented features; the precise choice of every printed time increment is not.

Full chart transcriptions and cell-by-cell comparison files are supplied electronically rather than repeated in the main text. The underlying 1981 fixture is unchanged in this revision. Source checksums, the exceptional use of the 1987 columns, and the calculation conventions are recorded alongside it.

2 Kinetics and calibration

The stated half-times are

$$(h_1, \dots, h_6) = (5, 10, 20, 40, 80, 120) \quad \text{minutes.}$$

The pressure convention recovered from the numerical table is an air inert-gas fraction $f = 0.79$ and surface absolute pressure of 33 feet of seawater (fsw). Thus $P_{\mathrm{s}} = 26.07$ is surface-equilibrium nitrogen pressure, and the equilibrium value at gauge depth d is $Q(d) = P_{\mathrm{s}} + fd$. Under exact half-time kinetics,

$$P_i(d,t) = P_{\mathrm{s}} + fd\bigl(1 - 2^{-t/h_i}\bigr). \tag{1}$$

Pressures in this note are fsw-equivalent nitrogen pressures unless stated otherwise. No water-vapor correction or timed descent/ascent is added to this historical square-exposure calculation.

2.1 The calibration is determined by column maxima

Let $\mathcal{E}$ be the twelve finite depth–time pairs printed in Table 1 of [1]. The reported construction is

$$\widehat{M}_i = \max_{(d,t)\in\mathcal{E}} P_i(d,t), \qquad M_i = \frac{\lfloor 2\widehat{M}_i \rfloor}{2}. \tag{2}$$

The downward half-unit rounding is stated explicitly [1, p. 5]. It reproduces all six published values (Table 2). Before rounding, $\widehat{M}$ is the coordinatewise least limit vector containing the supplied exposures: any such vector must dominate every column maximum. This elementary observation explains why rounding the maxima downward can exclude some of the original calibration endpoints.

Table 2: Calibration without fitted parameters. Exposure is depth/duration in fsw/minutes.

Half-time	Maximizing exposure	$\widehat{M_i}$	M_i
5	110/15	102.107500	102.0
10	100/20	85.320000	85.0
20	70/40	67.545000	67.5
40	50/75	54.801236	54.5
80	40/135	47.859323	47.5
120	30/225	43.308742	43.0

2.2 Arithmetic conventions and two discrepancies

Huggins writes the exponential coefficient as 0.693. Treating it as a printed approximation to $\log 2$ yields agreement in 70 of 72 pressure cells; using 0.693 literally yields agreement in 49. This supports, but does not historically prove, the exact half-time interpretation. The unavailable HP-67 program is not inferred from that comparison alone.

Decimal rounding also matters. At 70 fsw for 40 minutes, the 20-minute pressure is exactly 67.545; decimal half-up rounding gives the printed 67.55. Binary floating-point formatting can give 67.54. An independent 50-digit decimal evaluation avoids classifying this tie as an error.

The two remaining discrepancies are substantive:

Depth	Duration	Half-time	Printed	Calculated
35	165	40	52.41	52.14
60	50	120	39.96	37.96

Neither changes a column maximum. The numerical disagreements are established under (1); their explanation as printing or copying errors is a hypothesis, discussed in Section 6.

3 Group thresholds and residual times

The published upper group fractions are

$$\alpha_A = 0.63, \quad \alpha_B = 0.66, \quad \ldots, \quad \alpha_M = 0.99, \quad \alpha_N = 1.$$

These refer to total compartment pressure divided by M_i, not excess loading above surface equilibrium. The largest initial ratio is $P_\mathrm{s}/43 \approx 0.606279$, consistent with Huggins's reported 60.63% and the choice of 63% for group A [1, p. 5].

Define $R(d,t) = \max_i P_i(d,t)/M_i$. The first time this score reaches α_g is

$$T_g(d) = \min_i T_{i,g}(d), \qquad T_{i,g}(d) = \frac{h_i}{\log 2} \log \frac{Q(d) - P_\mathrm{s}}{Q(d) - \alpha_g M_i}, \tag{3}$$

when $P_\mathrm{s} < \alpha_g M_i < Q(d)$. A target at or above $Q(d)$ has infinite crossing time. All the stated group targets exceed P_s. Equation (3) is obtained by inverting (1), not by fitting the printed chart.

3.1 First-dive times

Of 127 numerical first-dive entries, 120 satisfy the associated continuous group bound. Seven group-N endpoints exceed it, at depths 30, 35, 40, 50, 70, 100, and 110 fsw. These entries retain

the empirical limits used in calibration. For example, the 30-fsw, 225-minute exposure produces 43.308742 in the 120-minute compartment, whereas its rounded limit is 43; the inverse model reaches that rounded limit at about 216.919 minutes. This is the consequence of combining independently retained endpoints with downward-rounded maxima, not a contradiction in the exponential formula.

Only 35 of the 127 numerical entries equal the integer floor of (3). The chart also uses nonuniform increments, arrows, and omissions. We preserve those choices without asserting a unique generator. The seven endpoint exceedances do not themselves settle consistency of subsequent surface intervals; those are checked directly in Section 5.

3.2 Residual-nitrogen-time candidate

The natural numerical candidate

$$R_g^{\mathrm{cand}}(d) = \lceil T_g(d) \rceil \tag{4}$$

agrees with 169 of the 179 finite printed residual times. Two printed dashes coincide with infinite candidates. The remaining dash, L at 20 fsw, has a finite candidate of 570 minutes. Table 3 gives every mismatch. The existence of a mismatch is established; its classification as an error or an undocumented convention is not.

Table 3: Disagreements with (4). A printed dash remains a dash.

Group	Depth (fsw)	Printed	Candidate (min)
L	20	dash	570
N	30	225	217
N	40	135	131
N	50	75	74
M	70	40	39
N	70	41	40
N	100	21	20
A	110	2	3
N	110	16	15
N	120	13	12
N	130	11	10

The first-crossing candidate is not a universal conservative residual-time theorem. It takes a *minimum* over compartments. A fictitious clean exposure of that duration need not dominate every compartment state compatible with the same group. Agreement with many chart entries does not establish full-state equivalence for arbitrary repetitive histories.

4 Surface intervals and their nearly additive structure

Let $L(g, j)$ be the printed lower surface-interval endpoint, in whole minutes, for departure group g and lower destination group j. There are 91 such strict transitions. The full chart has 105 ranges after its 14 same-group ranges are included.

4.1 A derived worst-case threshold

For an initial state satisfying $P_i(0) \le \alpha_g M_i$, ordinary half-time washout gives

$$P_i(s) = P_{\mathrm{s}} + (P_i(0) - P_{\mathrm{s}})2^{-s/h_i}.$$

The least time that guarantees $P_i(s) \le \alpha_j M_i$ for every state in that componentwise box is

$$B(g,j) = \max_i \frac{h_i}{\log 2} \log \frac{\alpha_g M_i - P_{\mathrm{s}}}{\alpha_j M_i - P_{\mathrm{s}}}. \tag{5}$$

Sharpness here concerns that box; it does not assert that every point of the box is attainable by a permitted exposure history.

Proposition 1 (Controlling surface compartment)**.** *For the stated parameters and all $g > j$, the 120-minute compartment attains the maximum in* (5)*.*

Proof. Put $z_i = P_{\mathrm{s}}/M_i$. Both h_i and z_i increase with i, and

$$\frac{d}{dz} \log \frac{\alpha_g - z}{\alpha_j - z} = \frac{\alpha_g - \alpha_j}{(\alpha_g - z)(\alpha_j - z)} > 0.$$

The logarithm is positive throughout the relevant range. Multiplication by the increasing positive h_i preserves the ordering. □

All 91 printed lower endpoints exceed $B(g,j)$, by between 3.187513 and 20.647090 minutes. None equals $\lceil B(g,j) \rceil$. For instance, B to A is printed as 151 minutes against 141.518844 from (5); N to A is printed as 507 against 486.352910. Thus ordinary rounding up to a minute is insufficient. We do not infer an undocumented safety adjustment merely from this inequality.

4.2 A cross-cell identity

The additional structure can be stated without identifying its historical cause. Use the destination-A row and departure-N column to define

$$\widehat{L}(g,j) = L(g,A) + L(N,j) - L(N,A). \tag{6}$$

Those row and column entries give 25 anchoring identities among the 91 transitions. Of the remaining 66 predictions, 63 are exact. The three exceptions are

Transition	Printed L	Predicted $\widehat{L}$	Residual
L to I	61	60	1
L to J	42	41	1
L to K	24	23	1

This is an integer identity checked against the fixture, not a statistical validation on an independent sample. All three exceptions involve departure group L. The pattern suggests departure- and arrival-dependent thresholds, but does not uniquely determine them.

What the identity does not prove. The form in (6) is a separate row offset plus column offset, not necessarily one common group potential. Indeed, the simplest proposed rule $L(g,j) = \lceil \tau_g - \tau_j + c \rceil$, with a single constant c, is incompatible with the printed entries. The triangle quantities

$$L(D,B) + L(B,A) - L(D,A) = 9, \qquad L(D,C) + L(C,A) - L(D,A) = 3$$

would both have to lie in $(c-1, c+2)$ under that rule: write each ceiling error as a number in $[0,1)$. Their separation of six rules this out. A richer scalar model with different departure and destination cutoffs remains a possible explanation; identifying a simple historical generator is left open.

5 An exact finite compatibility certificate

Because some first-dive endpoints exceed their nominal rounded group limits, the box argument alone cannot certify their subsequent intervals. A direct calculation does not require that intermediate assumption.

Proposition 2 (First-dive/surface-interval certificate)**.** *For each of the 127 numerical first-dive endpoints and each applicable printed surface-interval range, the endpoint exposure followed by the range's shortest duration leaves every compartment below the destination group's upper bound. This gives 998 profile–interval combinations and 5,988 inequalities. Their minimum certified pressure margin is at least*

$$\frac{26068979127}{429496729600} > 0.06069657 \quad \textit{fsw.}$$

This statement uses the two-witness chart transcription described in Section 1 and the square-exposure model (1)*.*

Proof. An exposure (d,t) followed by s minutes at the surface has pressure

$$P_i^{\text{after}} = P_{\text{s}} + fd\left(2^{-s/h_i} - 2^{-(s+t)/h_i}\right). \tag{7}$$

The data d, t, s, h_i, the group percentages, and all pressure constants are integers or rationals. For integer $n \geq 0$ and $h > 0$, enclose $x = 2^{-n/h}$ on the dyadic grid with denominator 2^{32}. If $32h \geq n$, find the largest integer m such that

$$m^h \leq 2^{32h-n}.$$

Integer binary search gives $m/2^{32} \leq x \leq (m+1)/2^{32}$, with equality at both bounds when the power equality is exact. If $32h < n$, use $0 \leq x \leq 2^{-32}$.

In (7), substitute an upper bound for the first decay factor and a lower bound for the second. The resulting pressure upper bound is rational. The program compares it exactly with the rational target $\alpha_j M_i$ in each of the 5,988 cases. All differences are positive; their least value is the displayed fraction. The checker and test assertions are supplied with the data. □

This is a computational proof of a finite arithmetic statement conditional on the transcribed chart, not a proof of physiology or historical intent. The formulas and the enclosure method are displayed so that the calculation can be repeated independently. No floating-point tolerance is used to decide these 5,988 inequalities.

For these initially equilibrated square exposures, shorter bottom durations give no larger pressures, and longer surface intervals decrease every loading. Hence the same upper-bound conclusions extend within the chart ranges bounded by the checked endpoints. This does not cover arbitrary repeated or multilevel exposures, nor does a twelve-hour chart cutoff imply exact return to equilibrium.

5.1 Irredundancy of the six fixed compartments

Each original compartment uniquely controls $T_N(d)$ at a benchmark depth:

Depth (fsw)	30	40	50	70	100	110
Controlling half-time (min)	120	80	40	20	10	5

Deleting any one of these fixed constraints therefore changes the continuous boundary. This does not prove a six-compartment minimum after refitting parameters, or for reproducing a finite rounded chart within a tolerance. Those are different optimization problems.

6 Explanations and open questions

The preceding calculations establish model formulas, numerical comparisons, an additive pattern, and a finite certificate. They do not establish why every historical chart entry was selected. The following interpretations are kept separate from those results.

Probable printing errors. The discrepancy 52.41 versus 52.14 is consistent with transposed digits. The erroneous 39.96 at 60 fsw is also the correct 120-minute value in the preceding 50-fsw row, making a copy-forward error plausible. These are hypotheses about production of the table. The original numbers remain in the fixture, and no contemporaneous erratum is asserted.

Preserved empirical limits. The coexistence of independently supplied Spencer endpoints and downward-rounded M-values directly explains the seven terminal model-bound exceedances. It does not, without further evidence, explain every residual-time exception. In particular, one-minute disagreements should not all be called either conservative adjustments or printing errors.

Historical scalar-cutoff hypothesis. The nearly additive surface chart is consistent with calculating in a scalar washout coordinate using distinct departure and destination cutoffs, followed by rounding or a small number of corrections. The cutoffs and corrections are not recovered here. The simpler single-potential, common-buffer rule has already been excluded above. This is a hypothesis about a missing algorithm, not an established property of the diver's physiological state.

Question 1: the surface-interval generator. Is there a simple, source-compatible rule for the departure and arrival cutoffs and integer rounding that reproduces every surface-interval entry? A successful formula should explain its parameters, not merely encode the table as fitted offsets. Recovery of the original HP-67 routines would supply independent evidence.

Question 2: display and residual-time conventions. Which grid-selection, blank-cell, terminal-time, or version-specific rules account for the residual-time mismatches and the nonuniform first-dive increments? The known rightward-arrow instruction answers one part of this question, not the entire chart-generation problem.

Question 3: minimum model dimension. What is the smallest number of compartments in a refitted model that reproduces a specified chart within one minute? The admissible parameter class, treatment of dashes and blanks, and the collection of first-dive and repetitive outputs must be fixed before a minimum is meaningful. Irredundancy of the six original constraints is only a starting point.

Question 4: empirical reconstruction. Can the original experimental Doppler observations, their grading protocol, and the selection of exposure limits reproduce the Spencer inputs independently? Neither the present numerical audit nor agreement with Huggins's formulas answers that statistical question.

7 Reproducibility and status

The repository separates source-preserving data, mathematical functions, candidate rules, and audit outputs. Full charts remain in the fixture; per-cell comparisons are generated rather than typeset as three large appendices. To reproduce the checks from the repository root:

```
python -m unittest discover -s tests -p 'test_huggins*.py' -v
python models/huggins_1981/huggins.py \
    --audit-dir validation/huggins_1981
python -m models.huggins_1981.surface_structure \
    --audit-dir validation/huggins_1981
sh papers/spencer_huggins/build.sh
```

The first command runs 19 Huggins-specific tests: the original 15 kinetic, source-preservation, inversion, and certificate checks, plus four integer-structure checks. Other repository tests are separate. The audit uses two distinct numerical paths for the threshold calculation and an independent high-precision decimal calculation for pressures. This is internal mathematical and numerical verification, not independent peer review or medical certification.

The source fixture, generated ledgers, code, and this LaTeX manuscript are versioned at `https://github.com/long-mathematics/dive-decompression-models`. The distribution manifest records file hashes. Original software is licensed under Apache-2.0 and original exposition under CC BY 4.0; neither policy relicenses the historical source material.

The resulting note is complete as an account of what has been reconstructed and checked. It deliberately leaves the undocumented choices as open questions rather than assigning them unsupported explanations.

References

[1] K. E. Huggins, *New No-Decompression Tables Based on No-Decompression Limits Determined by Doppler Ultrasonic Bubble Detection*, Michigan Sea Grant, MICHU-SG-81-205 (1981). NOAA Institutional Repository, `https://repository.library.noaa.gov/view/noaa/42176`. Printed pp. 1–6; PDF pp. 4–9.

[2] K. E. Huggins, *Microprocessor Applications to Multi-Level Air Decompression Problems*, Michigan Sea Grant, MICHU-SG-87-201 (1987). NOAA Institutional Repository, `https://repository.library.noaa.gov/view/noaa/40558`. Printed p. 13 and Appendix D, p. 99; PDF pp. 22 and 108.